



# Hybrid Nano-Modified Paraffin–Metal Foam System for Passive Thermal Management of Photovoltaic Modules

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**Abstract:** Elevated operating temperature remains a major constraint on the electrical performance of photovoltaic modules, particularly under sustained solar irradiation. Latent heat storage using phase change materials (PCMs) provides a passive means of controlling photovoltaic temperature ( $T_{PV}$ ), although the low thermal conductivity of conventional paraffin limits heat penetration and phase-change utilization. This study investigates a hybrid thermal management system that combines paraffin RT-25, Ag–Al<sub>2</sub>O<sub>3</sub>–TiO<sub>2</sub> ternary nanoparticles, and metal foam within a trapezoidal PCM enclosure beneath a photovoltaic module. A transient numerical model was developed to resolve the coupled heat transfer, PCM melting, and electrical response of the photovoltaic–PCM system. The thermophysical properties of the ternary nano-modified PCM were represented using a homogeneous formulation, and adaptive grid refinement was applied to resolve the evolving solid–liquid interface. The numerical model was also validated against published benchmark data. The combined use of ternary nanoparticles and metal foam increased the liquid fraction (LF) from 0.52 to 0.71 after 3 h, corresponding to an increase of approximately 35.65% relative to the pure-PCM configuration. Over the same period, the hybrid configuration reduced the  $T_{PV}$ , by approximately 3% and increased the electrical efficiency by about 5.25% relative to pure PCM cooling. Compared with the uncooled photovoltaic module, the complete hybrid system reduced the operating temperature from 363.15 to 331.04 K, a decrease of approximately 8.84%, while the electrical efficiency increased from 10.25% to 12.15%, corresponding to a relative improvement of approximately 18.55%. These results demonstrate that coupling a conductive porous network with ternary nano-modified paraffin promotes heat penetration and PCM melting while maintaining a lower photovoltaic operating temperature. The proposed configuration provides a passive thermal management approach for improving the thermal regulation and electrical performance of photovoltaic systems under sustained solar loading.

**Keywords:** Photovoltaic thermal management; Phase change material; Latent heat storage; Ternary nanoparticles; Metal foam; Photovoltaic electrical efficiency

## 1 Introduction

Improving the thermophysical response of phase change materials (PCMs) remains an important route for increasing the effectiveness of latent heat thermal energy storage and passive thermal regulation. A major limitation of conventional paraffin-based PCMs is their low thermal conductivity, which restricts heat penetration through the storage medium and slows the progression of the solid–liquid interface. Dispersing thermally conductive nanoparticles within the PCM has therefore been widely investigated as a means of modifying its effective thermal properties. Recent experimental studies and reviews of nano-enhanced PCM systems have shown that nanoparticle addition can accelerate thermal response and improve the performance of PCM-based photovoltaic thermal management [1, 2]. Ternary nanoparticle formulations extend this concept by combining three nanoparticle constituents with different thermophysical characteristics, providing additional flexibility in tailoring the effective properties of the storage medium.

Beyond nano-modification, incorporating porous metal foam into a PCM provides another effective means of addressing the limited thermal conductivity of paraffin. The interconnected metallic skeleton creates continuous high-conductivity pathways through the storage domain, improving temperature uniformity and accelerating phase-change progression [3]. Conductive internal structures can similarly promote heat penetration through PCM enclosures and

strengthen passive photovoltaic–PCM thermal regulation [4, 5]. Because nanoparticle modification acts primarily on the effective properties of the PCM while conductive structures provide extended heat-transfer pathways through the enclosure, combining the two mechanisms provides a physically plausible route for improving transient heat transfer at different spatial scales. Their coupled influence on phase-front development, latent heat utilization, and photovoltaic thermal regulation therefore warrants further investigation.

Solar photovoltaic systems convert only part of the incident solar radiation into electricity, while a substantial fraction is dissipated as heat. The resulting increase in cell temperature lowers electrical conversion efficiency and can also contribute to accelerated material degradation when modules operate for long periods under high solar loading. Thermal regulation is therefore closely related to both the energy performance and the long-term operation of photovoltaic systems. Photovoltaic–thermal concepts extend this principle by combining electrical generation with thermal management and, in some configurations, useful heat recovery. Maintaining a lower  $T_{PV}$ , improves electrical conversion while allowing part of the absorbed thermal energy to be transferred away from the module. For this reason, effective thermal management has become an important strategy for limiting photovoltaic operating temperature and maintaining electrical performance under sustained solar loading [6, 7]. The effectiveness of such systems depends strongly on the thermal unit coupled to the rear surface of the photovoltaic module. Conventional cooling approaches can remove heat rapidly, but their performance may depend on external power, fluid circulation, or continuously available heat sinks. PCM-based cooling provides a passive alternative because heat can be absorbed during the phase transition without the need for continuous mechanical operation. Its performance, however, is constrained by the low conductivity of the storage material. Introducing conductive structures or thermally modified PCMs can alleviate this limitation by promoting heat penetration, improving temperature uniformity, and extending the period over which latent heat contributes to  $T_{PV}$  regulation. Recent studies have shown that nano-enhanced PCMs and other heat-transfer modifications can improve the thermal response of photovoltaic and photovoltaic/thermal (PVT) systems and contribute to better electrical performance under sustained solar exposure [8–10].

A number of recent studies have demonstrated the value of PCM-assisted thermal management for photovoltaic applications. Elavarasan et al. [11] experimentally investigated a photovoltaic cooling arrangement combining PCM, fins, and water. The system increased daily electrical energy production by approximately 9.39% relative to the corresponding uncooled photovoltaic module. Li et al. [12] studied a building-integrated photovoltaic–PCM electric-film system intended for solar-assisted space heating and reported an operating efficiency of approximately 71.4%. Zheng et al. [13] developed a three-dimensional transient model for an aerogel-assisted photovoltaic–thermal–PCM configuration and examined its time-dependent thermal response and solar-energy utilization. Their optimized system increased overall efficiency by approximately 40.2% in summer and 42.4% in winter. Kibria et al. [14] assessed a PVT system incorporating paraffin modified with hybrid nanoparticles and reported an approximately 17.32% increase in overall system efficiency after the nano-modification was introduced. Sathish [15] investigated a solar-driven hydrogen production system coupling a photovoltaic electrolyzer, an evacuated-tube solar collector, and hybrid nanoparticle-modified PCM. Maximum electrolyzer and electrical efficiencies of approximately 39.2% and 15.5%, respectively, were reported. Collectively, these studies show that PCM-based thermal regulation can influence both  $T_{PV}$  and electrical performance, while also indicating that the effectiveness of the storage layer depends strongly on its internal heat-transfer characteristics.

Despite this progress, several issues remain unresolved in PCM-assisted photovoltaic thermal management. Pure paraffin can absorb a considerable amount of heat during melting, but slow thermal diffusion delays the movement of the phase interface and limits the fraction of the storage volume that can be effectively activated during a finite period of solar exposure. Metallic fins, porous structures, and nanoparticle additives have all been used to overcome this limitation. However, many previous investigations have focused on individual modification routes, whereas the combined influence of porous conductive networks and hybrid nanoparticles on transient PCM melting and the resulting photovoltaic response has received comparatively less attention. Porous metal structures can provide long-range conductive pathways and a large solid–PCM contact area, while nanoparticle addition modifies the effective thermophysical properties of the paraffin itself. The interaction between these mechanisms becomes particularly important when enclosure geometry, melting-front development,  $T_{PV}$ , and electrical efficiency are considered simultaneously. The coupled influence of geometric configuration, porous conduction, and ternary nano-modification on the transient thermal response of a photovoltaic–PCM system therefore requires further examination.

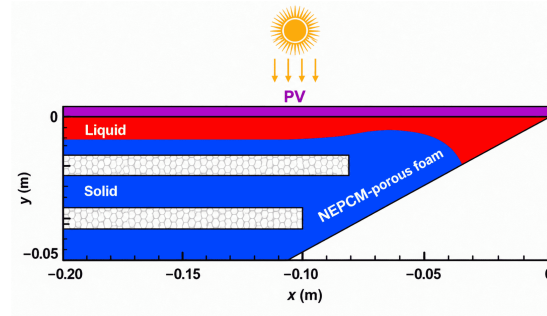
To address this issue, this study investigates a hybrid passive thermal management configuration in which a trapezoidal PCM enclosure is installed beneath a photovoltaic module and filled with RT-25 paraffin containing Ag–Al<sub>2</sub>O<sub>3</sub>–TiO<sub>2</sub> ternary nanoparticles and metal foam. The configuration combines material-level modification of the PCM with a continuous porous conductive network inside the storage domain. Rather than considering only the final module temperature, the analysis examines the evolution of the melting front, liquid fraction (LF), PCM temperature ( $T_{PCM}$ ), photovoltaic operating temperature, and electrical efficiency during transient solar loading. The numerical framework is used to determine how the two heat-transfer modification mechanisms affect phase-change progression and how these changes are subsequently reflected in the thermal and electrical response of the

photovoltaic module. The study therefore links internal PCM transport behavior with module-level photovoltaic performance, providing a basis for assessing hybrid latent-heat storage as a passive thermal management strategy for solar-energy applications.

## 2 Geometry and Modeling Assumptions of the Hybrid Phase Change Material System

The investigated system consists of a photovoltaic module coupled to a trapezoidal PCM enclosure positioned directly beneath its rear surface. Paraffin RT-25 is used as the base PCM because its phase-transition range is suitable for thermal regulation close to typical photovoltaic operating temperatures and because it provides latent heat storage within the temperature range considered in the present analysis. The inherently low thermal conductivity of paraffin, however, restricts heat penetration into the storage region and delays the melting process. To modify this behavior, ternary nanoparticles are assumed to be uniformly dispersed within the PCM, while metal foam is introduced throughout the paraffin domain as a continuous conductive structure. These two modifications serve different thermal functions: the nanoparticle mixture alters the effective thermophysical properties of the PCM, whereas the metal foam provides extended conductive pathways across the enclosure.

During operation, heat transferred from the photovoltaic module enters the PCM region through the upper interface of the storage unit. The combined presence of the nano-modified paraffin and metal foam is intended to redistribute this heat more rapidly through the enclosure, accelerate the progression of the solid–liquid interface, and increase the portion of the PCM involved in latent heat storage during the simulated period. The resulting thermal response is then reflected in the operating temperature and electrical efficiency of the photovoltaic module. For the ternary nano-modified PCM, the nanoparticles are assumed to remain uniformly dispersed and the mixture is represented using a single-phase homogeneous formulation. Under this assumption, the modified PCM is treated as an equivalent continuum with effective thermophysical properties determined from the constituent materials. This representation permits the thermal influence of the ternary nanoparticle mixture to be incorporated without resolving individual particle motion. The geometric arrangement of the photovoltaic module and trapezoidal PCM enclosure is shown in Figure 1.



**Figure 1.** Representation of the thermal management configuration consisting of a paraffin enclosure enhanced with ternary nanoparticles and metal foam for photovoltaic applications

Note: NEPCM = nano-enhanced phase change material; PV = photovoltaic.

Within the numerical framework, the photovoltaic module is treated as a thermal domain exchanging energy with both the ambient environment and the PCM enclosure located beneath it. The exposed photovoltaic surface loses heat to the surroundings through convection and thermal radiation, while heat conducted through the lower surface enters the PCM region and is stored through sensible heating and phase change. The transient temperature fields in the photovoltaic and PCM domains are coupled through the common interface. The governing energy equations and phase-change treatment follow the general numerical framework commonly adopted for photovoltaic–PCM thermal analysis [16–18]. The equations are adapted here to the present trapezoidal enclosure, ternary nano-modified paraffin, and metal-foam configuration, as described below.

$$\begin{aligned}
 (\rho_{PV} (C_p)_{PV}) \frac{\partial T_{PV}}{\partial t} &= k_{PV} \left( \frac{\partial^2 T_{PV}}{\partial y^2} + \frac{\partial^2 T_{PV}}{\partial x^2} \right) \\
 &+ \left( GA(1 - \eta_{PV}) \alpha_{PV} - \varepsilon_{PV} \sigma (T_{PV}^4 - T_{sky}^4) - h_w (T_{PV} - T_{\infty}) \right) \frac{1}{\delta_{PV}}, \\
 \eta_{PV} &= \eta_{ref} [1 - \beta_{ref} (T_{PV} - T_{ref})], \beta_{ref} = 0.0042(1/K), \eta_{ref} = 14.1\%
 \end{aligned} \tag{1}$$

$$(\gamma (\rho C_p)_{Tnf} + (1 - \gamma) (\rho C_p)_{GI}) \frac{\partial T_{PCM}}{\partial t} = (\gamma k_{Tnf} + (1 - \gamma) k_{GI}) \left( \frac{\partial^2 T_{PCM}}{\partial y^2} + \frac{\partial^2 T_{PCM}}{\partial x^2} \right) + (L\rho)_{Tnf} \frac{\partial S}{\partial t} \tag{2}$$

$$\begin{cases} T > (T_m + T_0) \Rightarrow & S = 0 \\ (-T_0 + T_m) < T < (T_0 + T_m) \Rightarrow & S = (-T + 0.5T_0 + T_m) / T_0 \\ T < (T_m - T_0) \Rightarrow & S = 1 \end{cases} \quad (3)$$

$$LF = 1 - S$$

where,  $t$  (s) denotes time, and  $x$  and  $y$  (m) represent the spatial coordinates. For the photovoltaic domain,  $T_{PV}$  (K) is the PV temperature,  $\rho_{PV}$  ( $\text{kg}\cdot\text{m}^{-3}$ ) and  $(C_p)_{PV}$  ( $\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$ ) are the density and specific heat capacity of the PV layer, respectively, and  $k_{PV}$  ( $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ ) is its thermal conductivity. The term  $\delta_{PV}$  (m) is the thickness of the PV layer. The absorbed solar radiation is characterized by the solar irradiance  $G$  ( $\text{W}\cdot\text{m}^{-2}$ ), the absorptivity  $\alpha_{PV}$  (dimensionless), and the packing factor  $A$  (dimensionless). The radiative heat loss to the sky is governed by the emissivity  $\varepsilon_{PV}$  (dimensionless), the Stefan–Boltzmann constant  $\sigma$  ( $\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-4}$ ), and the sky temperature  $T_{sky}$  (K), while convective heat loss is determined by the convective heat-transfer coefficient  $h_w$  ( $\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$ ) and the ambient temperature  $T_\infty$  (K). The electrical efficiency of the PV module,  $\eta_{PV}$  (dimensionless), is evaluated based on the reference efficiency  $\eta_{ref}$  and the temperature coefficient  $\beta_{ref}$  ( $\text{K}^{-1}$ ), as defined in Eq. (1), and  $T_{ref}$  is the reference temperature of the photovoltaic module.

For the PCM domain,  $T_{PCM}$  (K) is the local temperature;  $\gamma$  (dimensionless) is the volume fraction of the nano-enhanced PCM within the composite domain;  $L$  ( $\text{J}\cdot\text{kg}^{-1}$ ) is the latent heat of fusion;  $T_m$  (K) is the melting temperature; and  $T_0$  (K) is the half-width of the phase-change temperature interval. The subscripts Tnf and GI refer to the ternary nanofluid (nano-enhanced PCM) and the gypsum layer (metal foam), respectively. The liquid fraction  $LF$  (dimensionless), defined as  $1 - S$ , serves as a direct indicator of the melting progress.

Paraffin RT-25 is adopted as the PCM, and the material properties used in the simulations are taken from the work of Biwole et al. [19]. Because the thermal conductivity of pure RT-25 is relatively low, Ag–Al<sub>2</sub>O<sub>3</sub>–TiO<sub>2</sub> ternary nanoparticles are incorporated to form a nano-modified PCM with modified effective thermophysical properties. The effective properties of the resulting nano-enhanced PCM are evaluated using mixture-based constitutive relations, following the approaches reported by Bilal et al. [20] and Nagaraja et al. [21].

$$\frac{k_{Thnf}}{k_{hnf}} = \frac{k_{s1} + 2k_{hf} - 2\phi_1 (k_{hf} - k_{s1})}{k_{s1} + 2k_{hf} + \phi_1 (k_{hf} - k_{s1})} \quad (4)$$

$$(\rho C_p)_{Thnf} = (1 - \phi_1) \left[ (1 - \phi_2) (\phi_3 (\rho C_p)_{s3}) + (\rho C_p)_{s2} \phi_2 + (1 - \phi_3) (\rho C_p)_f \right] + (\rho C_p)_{s1} \phi_1 \quad (5)$$

$$\frac{k_{hnf}}{k_{nf}} = \frac{k_{s2} + 2k_{nf} - 2\phi_2 (k_{nf} - k_{s2})}{k_{s2} + 2k_{nf} + \phi_2 (k_{nf} - k_{s2})} \quad (6)$$

$$\frac{k_{nf}}{k_f} = \frac{k_{s3} + 2k_f - 2\phi_3 (k_f - k_{s3})}{k_{s3} + 2k_f + \phi_3 (k_f - k_{s3})} \quad (7)$$

$$\rho_{Thnf} = [(1 - \phi_1) [(1 - \phi_2) ((1 - \phi_3) (\phi_3 \rho_{s3} + \rho_f)) + (\phi_2 \rho_{s2})] + \phi_1 \rho_{s1}] \quad (8)$$

$$(\rho L)_{Thnf} = (\rho L)_f (1 - \phi_1) (1 - \phi_2) (1 - \phi_3) \quad (9)$$

where the subscripts f,  $s_1$ ,  $s_2$  and  $s_3$  refer to the base fluid (pure PCM) and the three distinct types of solid nanoparticles (Ag, Al<sub>2</sub>O<sub>3</sub>, and TiO<sub>2</sub>), respectively. The parameters  $\phi_1$ ,  $\phi_2$  and  $\phi_3$  (dimensionless) denote the volume fractions of the corresponding nanoparticles. The subscript Thnf denotes the ternary hybrid nanofluid. The terms  $k_{nf}$  and  $k_{hnf}$  ( $\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ ) represent the intermediate effective thermal conductivities of the nanofluid and hybrid nanofluid, calculated sequentially during the multi-step homogenization process. Eqs. (4)–(9) define the effective thermal conductivity, volumetric heat capacity, and effective density and latent heat density of the resulting ternary nano-enhanced PCM.

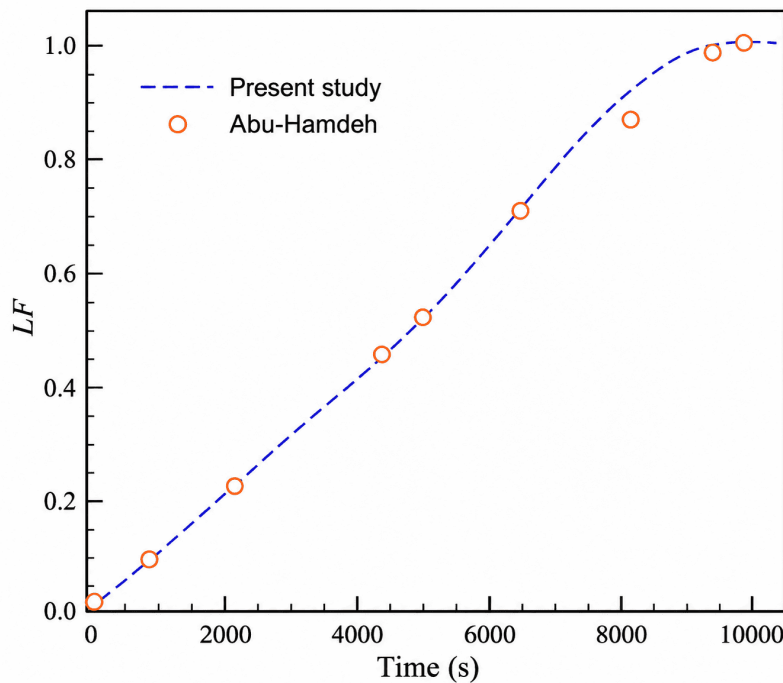
The photovoltaic–PCM system is subjected to a constant solar irradiance of  $650 \text{ W/m}^2$ , while an external convective heat-transfer coefficient of  $10 \text{ W/m}^2\cdot\text{K}$  is applied at the exposed photovoltaic surface. The lateral boundaries of the computational domain are assumed to be adiabatic, whereas the upper surface of the photovoltaic module exchanges heat with the surroundings through convection and thermal radiation. Transient melting is simulated over a period of 3 h using FLEXPDE. The governing equations are solved using the Galerkin finite-element formulation, with adaptive grid refinement applied to increase spatial resolution in regions of steep temperature gradients and near the evolving solid–liquid interface. This treatment allows the computational mesh to follow the transient phase-change region without imposing uniformly fine resolution throughout the domain. A related numerical implementation for nanoparticle-enhanced paraffin in photovoltaic thermal management is reported in Sheikholeslami's work [18].

### 3 Results and Discussion

The numerical analysis focuses on the transient interaction between PCM melting and photovoltaic thermal regulation and on the extent to which the combined use of ternary nanoparticles and metal foam modifies this interaction. Three configurations are considered in the performance comparison: an uncooled photovoltaic module, a photovoltaic module coupled to pure RT-25 paraffin, and the hybrid configuration incorporating nano-modified RT-25 and metal foam. The discussion first examines the reliability of the numerical solution and subsequently considers the evolution of the melting front, the coupled thermal response of the PCM and photovoltaic module, and the resulting electrical performance.

#### 3.1 Model Validation and Adaptive Mesh Refinement

To assess the underlying phase-change formulation independently of the proposed hybrid geometry, the numerical model was first validated against published liquid-fraction data for transient paraffin melting [22]. Figure 2 compares the predicted evolution of the LF with the corresponding reference results. The present predictions closely followed the benchmark data throughout the melting process, including the initial stage and the subsequent increase in LF toward complete melting. This agreement supports the ability of the adopted phase-change formulation to reproduce the transient melting behaviour required for the subsequent photovoltaic–PCM simulations.

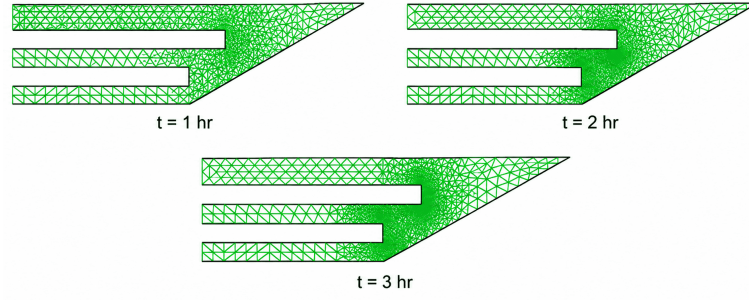


**Figure 2.** Validation of the predicted liquid fraction (LF) against published benchmark data [22]

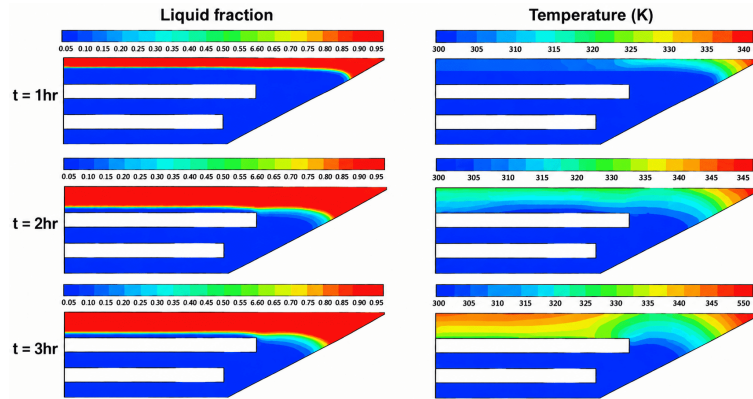
Accurate resolution of the moving solid–liquid interface is particularly important because the temperature gradient and phase fraction can vary sharply across this region. Figure 3 shows the adaptive computational mesh at different stages of the melting process. Rather than maintaining a uniformly dense mesh throughout the entire domain, the numerical procedure concentrated smaller elements in regions where the thermal field changed rapidly and around the advancing phase boundary. As melting progressed, the refined region moved with the developing interface. This adaptive treatment provided additional spatial resolution where it was required while avoiding unnecessary refinement in regions characterized by relatively smooth temperature variations.

#### 3.2 Transient Melting Behavior of the Phase Change Material

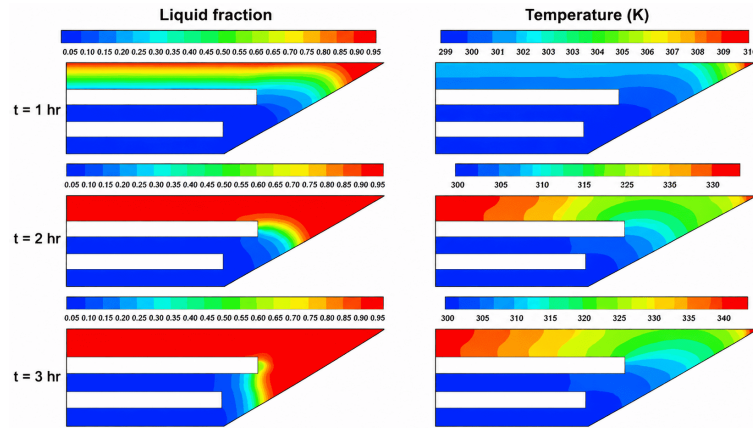
Figure 4–Figure 6 show the evolution of the temperature field, LF, and melting front during the 3 h simulation. For the pure-PCM configuration, heat entering the enclosure from the photovoltaic side initially produced melting close to the heated boundary. The low thermal conductivity of RT-25 restricted heat penetration toward the deeper parts of the enclosure, and the solid–liquid interface therefore advanced progressively through the PCM domain. Despite this limitation, the latent heat absorbed by the paraffin moderated the thermal response of the photovoltaic module. Relative to the uncooled configuration, pure PCM cooling reduced the photovoltaic operating temperature by approximately 6.02% and increased the electrical efficiency by about 12.62%.



**Figure 3.** Grid arrangement within the storage domain during the melting stage



**Figure 4.** Transient melting behavior of the phase change material (PCM) enclosure in the absence of thermal enhancement techniques

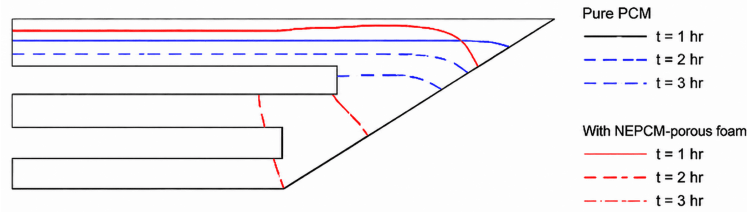


**Figure 5.** Transient melting behavior of the phase change material (PCM) enclosure in the presence of thermal enhancement techniques

The phase-change pattern changed when ternary nanoparticles and metal foam were incorporated into the storage medium. The metal foam provided continuous conductive paths extending from the heated region into the PCM volume, while the nano-modified paraffin exhibited higher effective thermal transport than the unmodified material. Their combined action redistributed heat through a larger portion of the enclosure and accelerated the movement of the melting front. This behavior is visible in Figure 5 and Figure 6, where the phase boundary penetrated farther into the lower part of the storage domain than in the pure-PCM case.

After 3 h, the LF increased from 0.52 for pure PCM to 0.71 for the hybrid configuration, corresponding to an increase of approximately 35.65%. In contrast, the  $T_{PCM}$  increased only slightly, by approximately 0.62%. The simultaneous increase in LF and comparatively small change in  $T_{PCM}$  is important because it indicates that the improved internal heat transport primarily promoted phase transition rather than simply producing a large rise in sensible temperature. The porous conductive network therefore allowed a greater portion of the PCM volume to

participate in latent heat storage within the same operating period. This response explains why modification of the internal transport process can influence  $T_{PV}$  even when the total amount of PCM remains unchanged.

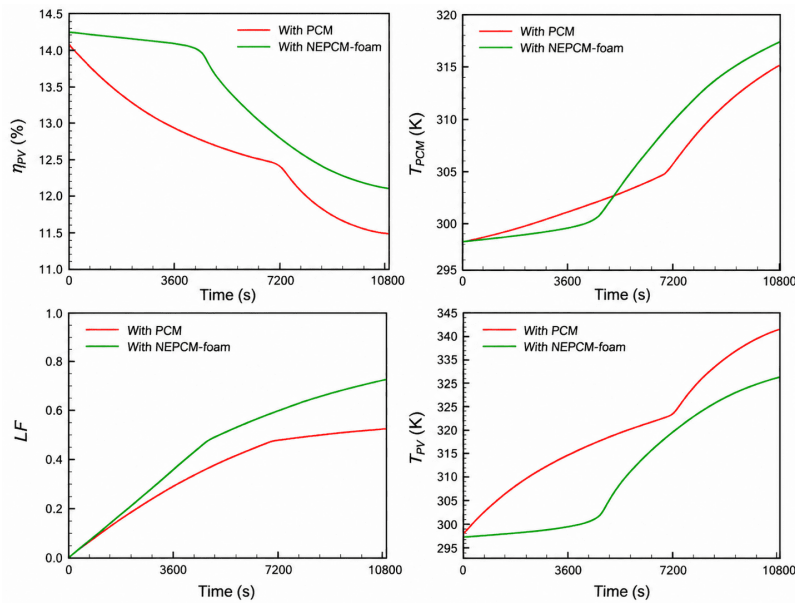


**Figure 6.** Effect of the applied enhancement approaches on the evolution of the melting front

Note: NEPCM = nano-enhanced phase change material; PCM = phase change material.

### 3.3 Coupled Thermal and Electrical Response

Figure 7 presents the temporal variations in  $T_{PV}$ ,  $T_{PCM}$ , LF, and photovoltaic electrical efficiency  $\eta_{PV}$ . Continuous solar irradiation caused heat to accumulate in the photovoltaic module, producing a progressive increase in  $T_{PV}$  during the simulation. Because photovoltaic electrical efficiency is temperature dependent,  $\eta_{PV}$  decreased as the module became warmer. At the same time, heat transferred from the photovoltaic module into the underlying storage layer increased  $T_{PCM}$  and progressively converted solid paraffin into liquid PCM.



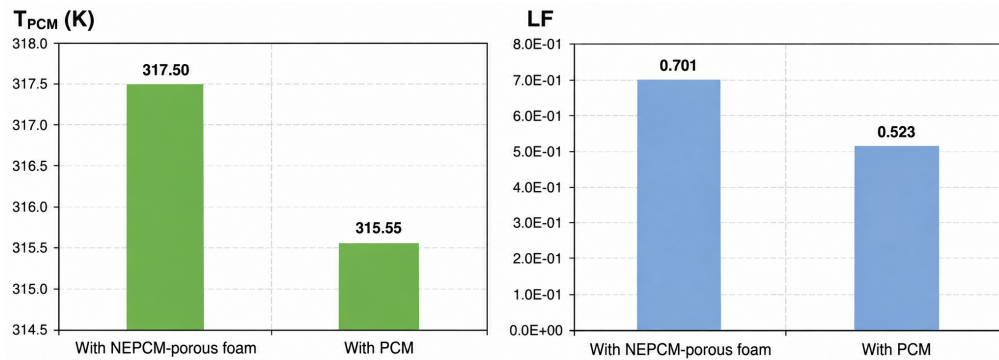
**Figure 7.** Temporal evolution of  $T_{PV}$ ,  $T_{PCM}$ , liquid fraction (LF), and  $\eta_{PV}$  during the melting process

Note: NEPCM = nano-enhanced phase change material; PCM = phase change material.

The temporal curves also reveal the role of internal PCM heat transfer in controlling the photovoltaic response. In the hybrid configuration, heat was transported away from the photovoltaic–PCM interface more rapidly and distributed over a larger storage volume. The corresponding increase in LF indicates that the additional heat penetration activated more of the available latent heat capacity. As a result, heat accumulation at the photovoltaic side was reduced and the increase in module temperature was moderated. The higher  $\eta_{PV}$  obtained with the hybrid system is therefore not an independent effect of the added materials; it follows from the altered thermal pathway between the photovoltaic module and the latent heat storage region.

Figure 8 provides a direct comparison between pure paraffin and the hybrid nano-modified PCM–metal foam configuration. The LF increased by approximately 35.65% after the conductive modifications were introduced, whereas  $T_{PCM}$  changed from 315.55 to 317.52 K. Over the same comparison,  $\eta_{PV}$  increased by approximately 5.25% and  $T_{PV}$  decreased by about 3%. These results establish a direct link between the accelerated phase transition inside the enclosure and the thermal regulation of the photovoltaic module. In particular, the relatively modest variation

in average PCM temperature compared with the substantial increase in LF suggests that the hybrid configuration altered how the absorbed thermal energy was distributed between sensible heating and phase change.

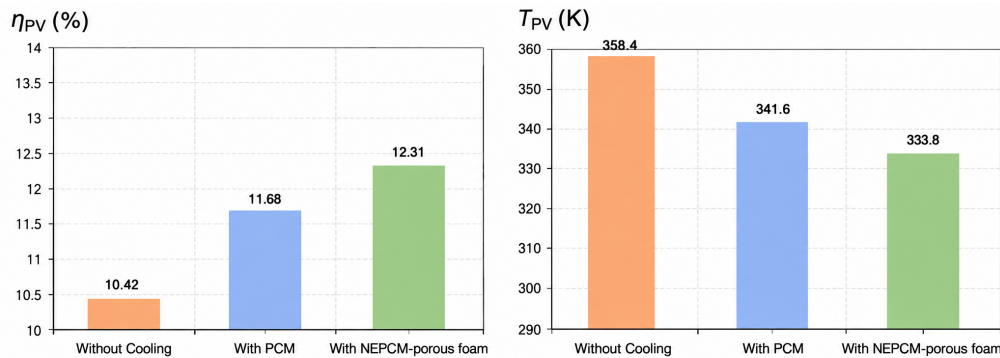


**Figure 8.** Comparison of phase change material (PCM) melting characteristics for the pure-PCM and hybrid nano-modified PCM–metal foam configurations

Note: NEPCM = nano-enhanced phase change material; LF = liquid fraction.

### 3.4 Performance Comparison of the Thermal Management Configurations

Figure 9 compares the photovoltaic operating temperature and electrical efficiency after 3 h for the three thermal management conditions. Without cooling, the photovoltaic module reached approximately 363.15 K and its electrical efficiency decreased to 10.25%. Adding a pure RT-25 layer reduced the module temperature to 341.29 K and increased  $\eta_{PV}$  to 11.54%. The latent heat capacity of the PCM therefore provided substantial passive temperature regulation even without additional conductive modification.



**Figure 9.** Effect of the thermal management configurations on photovoltaic module performance after 3 h

The hybrid configuration produced a further reduction in  $T_{PV}$  to 331.04 K, while the electrical efficiency reached 12.15%. Relative to the uncooled module, these values correspond to an approximately 8.84% reduction in operating temperature and an 18.55% relative increase in electrical efficiency. Relative to pure PCM cooling, the additional reductions in  $T_{PV}$  and increases in  $\eta_{PV}$  were approximately 3% and 5.25%, respectively. Distinguishing these two reference cases is important: the larger values quantify the overall effect of the complete thermal management system compared with an uncooled photovoltaic module, whereas the smaller values isolate the additional contribution obtained by modifying the PCM with ternary nanoparticles and metal foam.

The comparison also shows that the performance gain cannot be attributed solely to the latent heat capacity of paraffin. Pure RT-25 provided the storage capacity required for passive cooling, but its low thermal conductivity limited the rate at which heat could be transferred from the photovoltaic-facing region into the remaining PCM volume. The hybrid configuration addressed this transport limitation by combining material-scale modification of the PCM with a continuous porous conduction network. The resulting increase in melting progression allowed more of the storage medium to participate in heat absorption during the finite 3 h operating period and reduced thermal accumulation at the photovoltaic module.

From an energy-system perspective, the principal advantage of the proposed configuration is that  $T_{PV}$  regulation is achieved through passive latent heat storage rather than continuous mechanical cooling. The present results

therefore identify internal PCM heat transport as an important design variable for maintaining photovoltaic electrical performance under sustained solar loading. At the same time, the findings are limited to the specified numerical conditions, including a solar irradiance of 650 W/m<sup>2</sup>, the adopted convective boundary condition, and a 3 h operating period. Assessment under time-varying weather conditions, repeated melting–solidification cycles, and experimental operating conditions is required before the performance of the configuration can be generalized to long-term outdoor photovoltaic applications.

#### 4 Conclusions

A transient numerical investigation was conducted to examine passive thermal management of a photovoltaic module using a trapezoidal enclosure containing RT-25 paraffin, Ag–Al<sub>2</sub>O<sub>3</sub>–TiO<sub>2</sub> ternary nanoparticles, and metal foam. The analysis linked the internal phase-change behavior of the storage medium to the thermal and electrical response of the photovoltaic module and compared the hybrid configuration with pure PCM cooling and an uncooled photovoltaic module.

The principal findings are as follows:

1. Pure RT-25 provided effective passive thermal regulation through latent heat absorption. After 3 h, the photovoltaic operating temperature was approximately 6.02% lower than that of the uncooled module, while the electrical efficiency was approximately 12.62% higher.
2. Combining ternary nanoparticles with metal foam accelerated heat penetration through the PCM enclosure and promoted the progression of the melting front. The LF increased from 0.52 to 0.71 relative to the pure-PCM case, corresponding to an increase of approximately 35.65%, while the PCM temperature changed only moderately.
3. The accelerated phase transition translated into additional photovoltaic thermal regulation. Compared with pure PCM cooling, the hybrid configuration reduced the photovoltaic operating temperature by approximately 3% and increased electrical efficiency by about 5.25%.
4. The complete hybrid system reduced the  $T_{PV}$  from 363.15 K in the uncooled case to 331.04 K and increased the electrical efficiency from 10.25% to 12.15%. These changes correspond to an approximately 8.84% reduction in operating temperature and an 18.55% relative increase in electrical efficiency.

The results demonstrate that coupling nano-modified paraffin with a porous conductive network can improve the use of latent heat storage during a finite period of solar exposure and thereby moderate temperature-induced losses in photovoltaic conversion. The proposed configuration provides a passive approach to photovoltaic thermal management without relying on continuous mechanical cooling. Future work should examine the system under variable solar irradiance and ambient conditions, repeated thermal cycles, and experimental operation to determine its long-term thermal response and practical applicability.

#### Author Contributions

Conceptualization, M.H.A.; supervision, M.H.A.; software, M.H.A.; writing—original draft preparation, M.H.A.; investigation, F.A.M.A.A.; visualization, F.A.M.A.A.; writing—review and editing, F.A.M.A.A.; validation, F.A.M.A.A. All authors have read and agreed to the published version of the manuscript.

#### Data Availability

The data used to support the research findings are available from the corresponding author upon request.

#### Conflicts of Interest

The authors declare no conflicts of interest.

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